



## Basic Principles of Medicine 1

### Module: Musculoskeletal System | IMCQ1

# IMCQ 1

*School of Medicine,  
St. George's University*

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**1. A 29-year-old man comes to the clinic due to a 2-month history of left-sided shoulder pain. He describes the pain as deep and localized near the lower portion of his scapula. Physical examination shows localized tenderness on deep palpation of the inferior aspect of the left scapula but no muscle wasting or atrophy. Both passive and active ranges of motion of the shoulder are normal. Needle electromyography is planned to assess for possible Dorsal scapular nerve damage. Which of the following muscles would most likely be tested to evaluate for this nerve dysfunction in this patient?**

- A. Rhomboids and levator scapulae
- B. Levator scapulae and trapezius
- C. Rhomboids and infraspinatus
- D. Supraspinatus and infraspinatus
- E. Subscapularis and trapezius

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**2. A 45-year-old man comes to the physician because of a 3-day history of lower back pain that radiates down the back of the right leg. Physical examination shows no tenderness of the paravertebral muscles, but there is weakness in the right foot. An MRI of the patient is shown below. Which of the following is the most likely level of the injury?**

- A. L1-L2
- B. L2-L3
- C. L3-L4
- D. L4-L5
- E. L5-S1



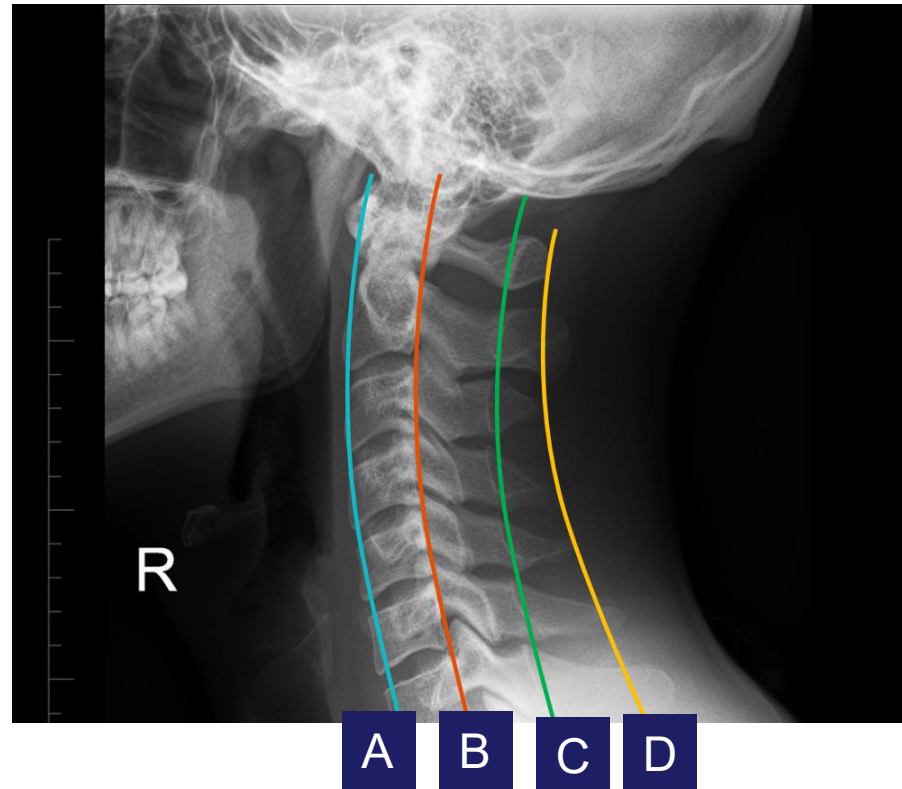
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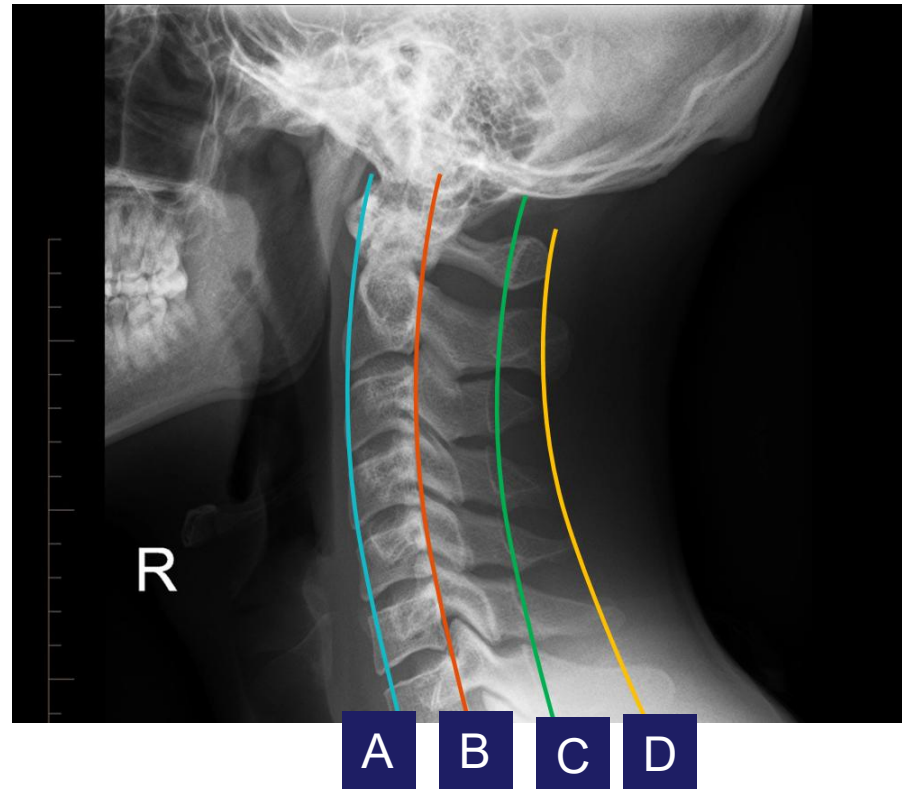
3. An 18-year-old man is brought to the emergency department following a high-speed motor vehicle collision. He was a restrained front-seat passenger in a car that was rear-ended on a major highway. On arrival, he is conscious but in severe pain, and his neck is immobilized in a cervical collar. Physical examination shows tenderness over the cervical spine but no focal neurological deficits. Hyperextension injury of the cervical spine with a torn ligament is suspected. In the attached radiograph, which of the labeled lines most likely represents the location of the torn ligament in this patient?

- A. A
- B. B
- C. C
- D. D



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- B. ☐ B
- C. ☐ C
- D. ☐ D



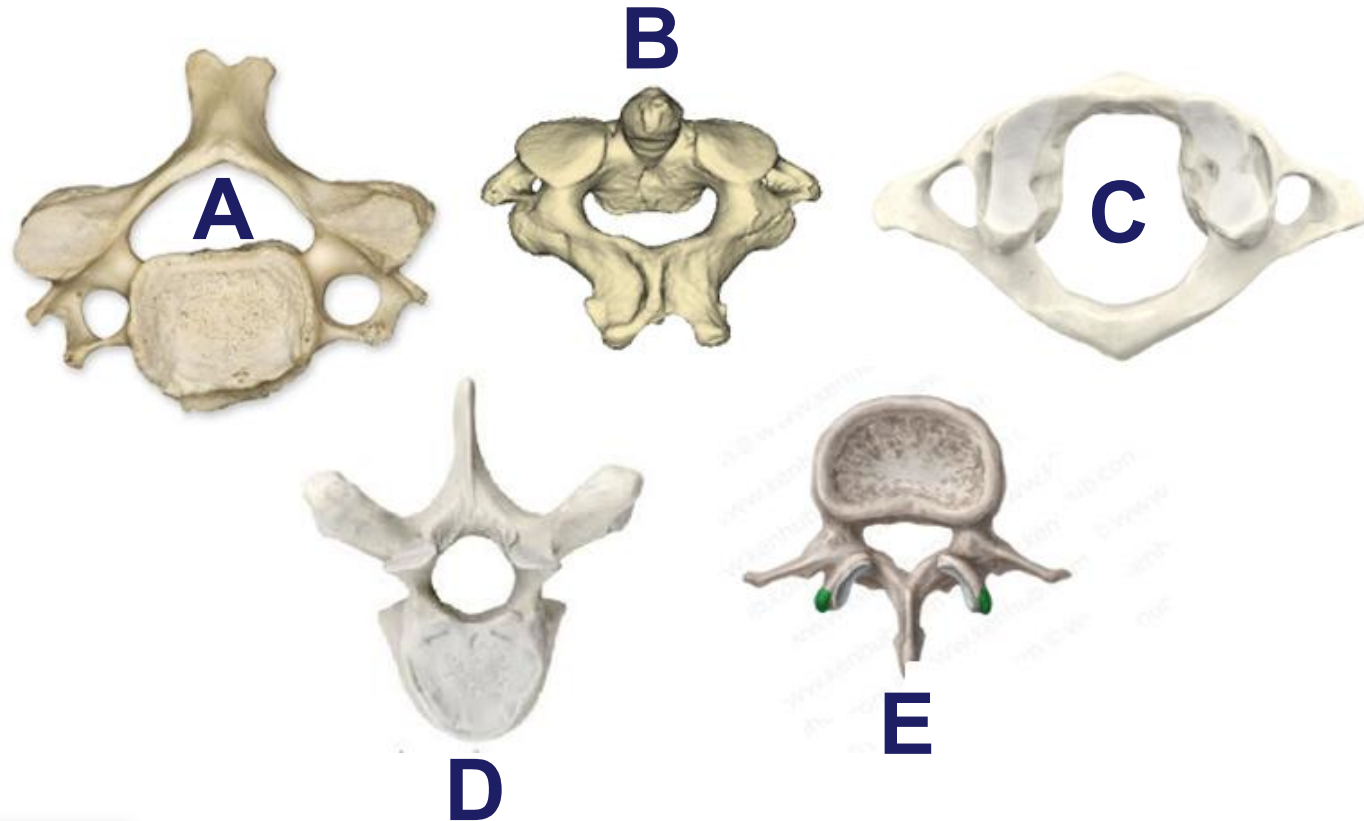
SOM.MKII.BPM1.1.  
MSK.1.ANAT.1049

Describe the five ligaments  
that support the vertebral  
column



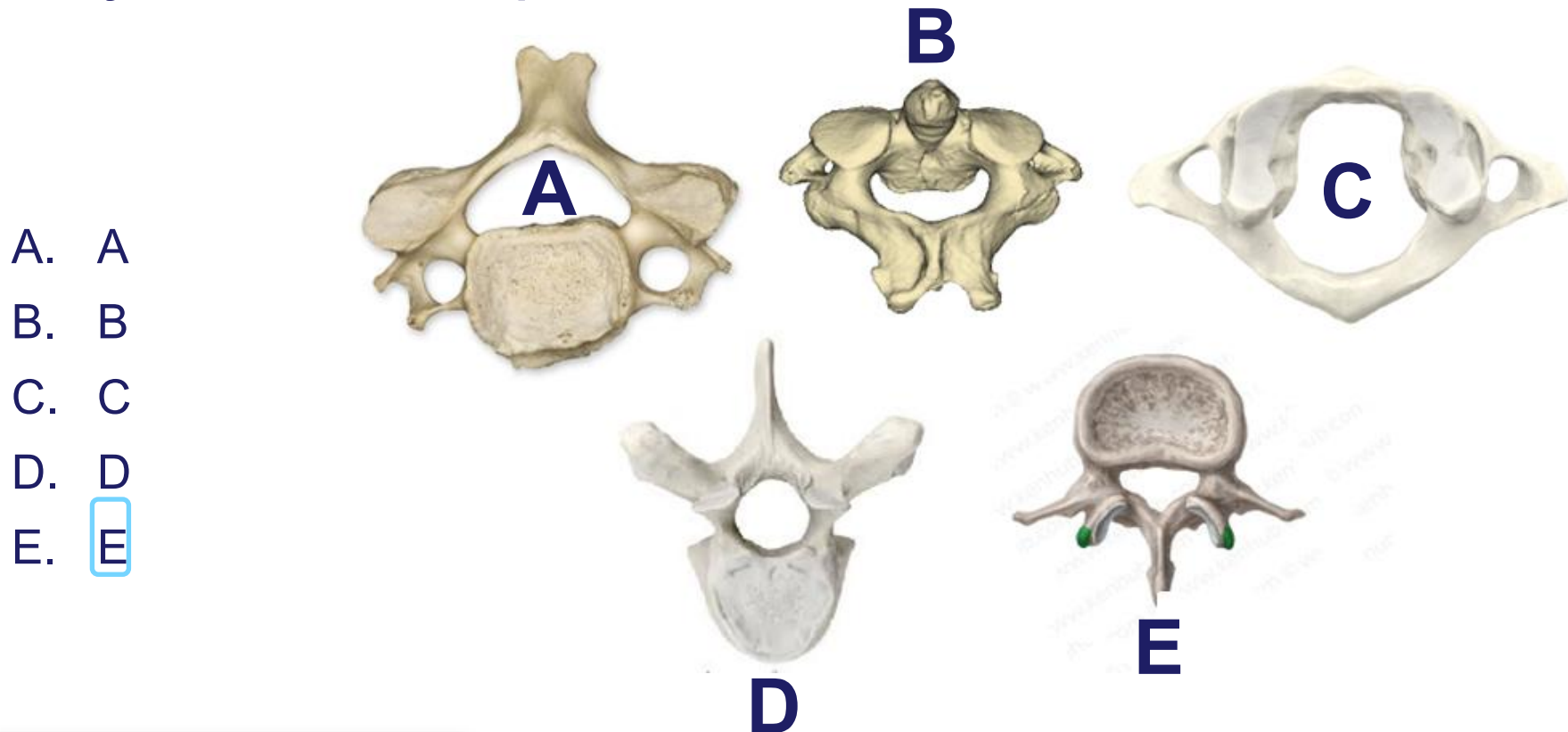
4. A 36-year-old man is brought to the emergency department after falling from a 10-foot ladder 2 hours ago. He landed on his feet but immediately felt severe lower back pain. Physical examination shows point tenderness over the lower back and limited spinal range of motion. A CT scan of the spine shows a vertebral compression fracture. Which of the following vertebrae is most likely fractured in this patient?

- A. A
- B. B
- C. C
- D. D
- E. E





4. A 36-year-old man is brought to the emergency department after falling from a 10-foot ladder 2 hours ago. He landed on his feet but immediately felt severe lower back pain. Physical examination shows point tenderness over the lower back and limited spinal range of motion. A CT scan of the spine shows a vertebral compression fracture. Which of the following vertebrae is most likely fractured in this patient?



- A. A
- B. B
- C. C
- D. D
- E. E

5. A 28-year-old man comes to the physician because of a 2-week history of right-sided neck pain and tingling in his right index and middle fingers. Pain worsens with neck extension. Physical exam reveals weakness in right wrist extension, reduced triceps reflex on the right. A cervical MRI is shown. Which of the following nerves would most likely be compressed in this patient?

- A. C4
- B. C5
- C. C6
- D. C7
- E. C8



Witiw CD, Mathieu F, Nouri A, Fehlings MG. Clinico-Radiographic Discordance: An Evidence-Based Commentary on the Management of Degenerative Cervical Spinal Cord Compression in the Absence of Symptoms or With Only Mild Symptoms of Myelopathy. *Global Spine Journal*. 2018;8(5):527-534. doi:10.1177/2192568217745519

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6. A 63-year-old woman comes to the physician because of a 2-week history of neck pain. The pain started after the patient fell asleep in her chair hitting the right side of her neck and shoulder. Initially, she had minimal neck pain and was evaluated by her primary care physician, which showed no neurological signs and she was sent home on pain medication. Over the last two weeks, she has developed numbness in both of her upper extremities. Physical examination shows bilateral hyper-reflexia in the upper extremities. Computed tomography of the cervical spine was performed upon arrival for better delineation of the bony injury with confirmation of fracture of the structure indicated by the arrow in the X-ray below. Which of the following ligaments is most likely closely associated with the pointed structure and helps to prevent its displacement?



- A. Alar ligament
- B. Apical ligament
- C. Nuchal ligament
- D. Ligamentum flavum
- E. Transverse ligament



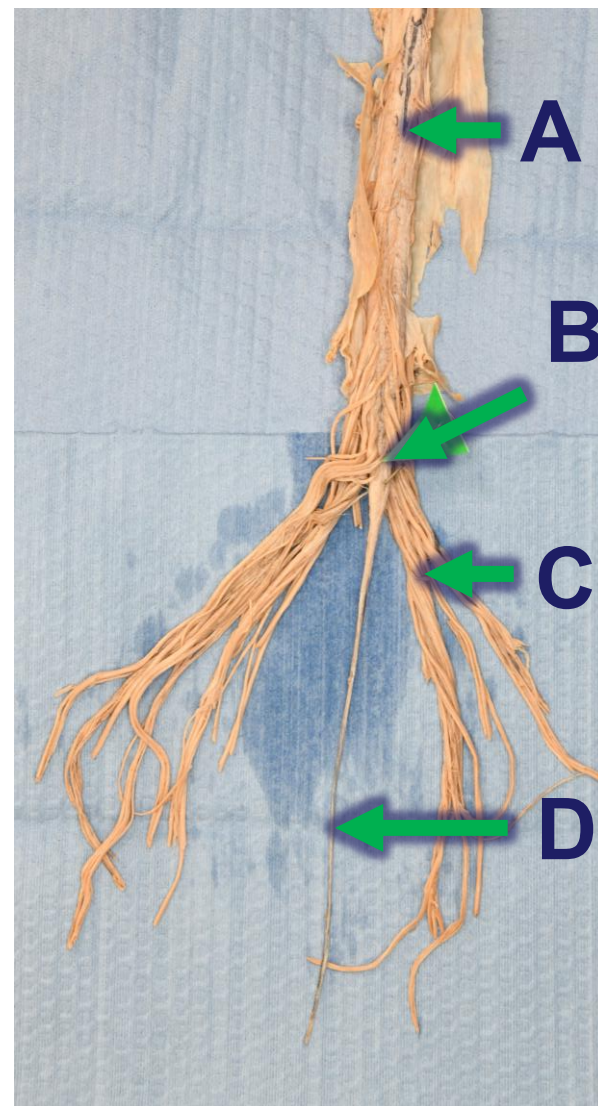
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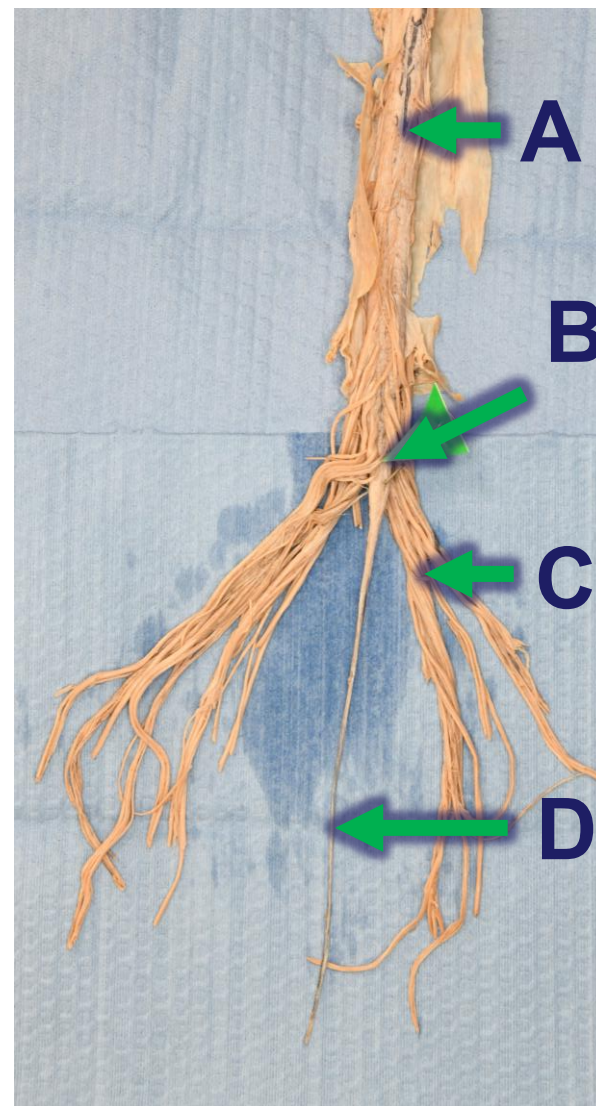
**7. A 54-year-old man comes to the emergency department because of a 2-day history of severe lower back pain and urinary retention. He also reports numbness and tingling in both legs. His medical history is significant for long-standing degenerative disc disease. Physical examination shows bilateral lower extremity weakness, hyperreflexia, and decreased sensation in a saddle-shaped distribution. An MRI of the lumbar spine shows a posterior herniation of L1/L2 intervertebral disc. Which of the following structures depicted in the attached cadaveric photograph is most likely injured in this patient?**

- A. A
- B. B
- C. C
- D. D



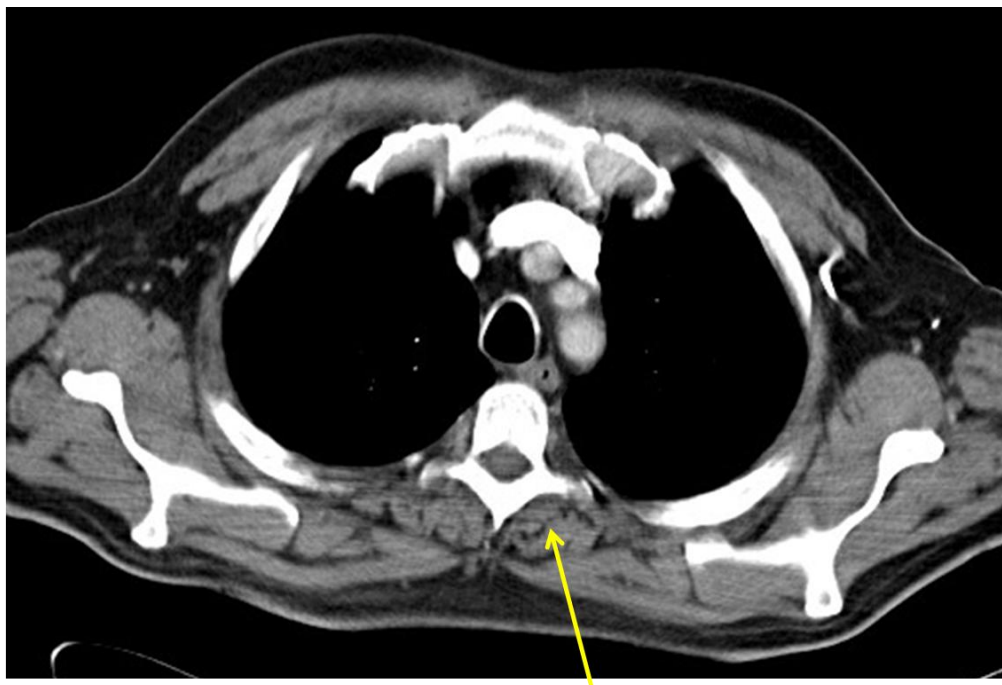
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- B. **B**
- C. C
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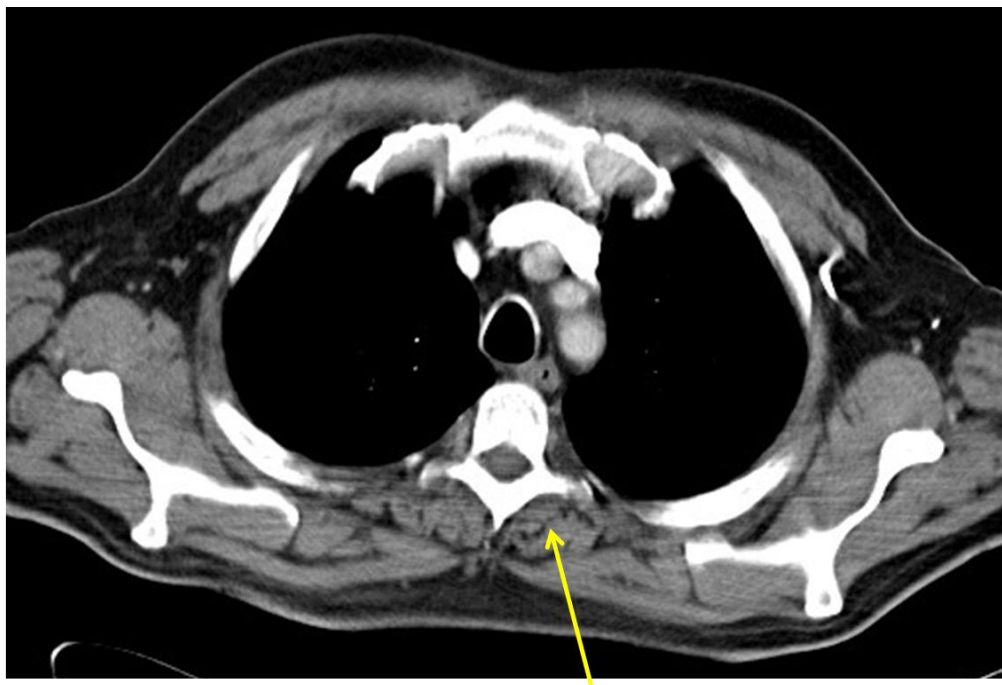


8. A 24-year-old man comes to the physician because of a 10- day history of lower back pain and periodic spasms. He is a weightlifter and is preparing for a competition by increasing his training hours and intensity over the last month. Physical examination shows paraspinal tenderness in the right lumbar region. Passive range of motion of the vertebral column is normal though active extension is somewhat limited by pain. Neurological testing shows no deficit. The physician suspects a sprain affecting the muscle pointed out in the image. Which of the following statements is the best description of this muscle?



- A. It is located superficial to the thoracolumbar fascia
- B. Its primary action is flexion of the vertebral column
- C. It is innervated by branches of the dorsal rami
- D. Its primary action is elevation of the ribs
- E. It is innervated by branches of the brachial plexus
- F. It forms the medial boundary of the lumbar triangle

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**9. A 25-year-old man is brought to the emergency department because following a stab wound to the back 2 hours ago. Physical examination shows active bleeding from a 5cm deep laceration to the inferolateral region of his left back. He has difficulty extending adducting and medially rotating the arm. The muscle affected in this patient most likely receives arterial blood supply from which of the following vessels?**

- A. Intercostal
- B. Thoracodorsal
- C. Dorsal scapular
- D. Transverse cervical
- E. Lateral sacral

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**10. A 64-year-old man is brought to the emergency department because of a 4-day history of painful rash. His pulse is 108/min, respirations are 16/min, blood pressure is 113/88 mm Hg, and temperature is 41°C (106°F). Physical examination shows skin eruptions that are localized entirely on the right side of his abdomen, closely following the skin lateral to the umbilicus. The patient was diagnosed with a herpes zoster viral infection known as “shingles.” Which of the dermatomes is most likely involved in this patient?**

- A. C6
- B. T2
- C. T7
- D. T10
- E. L2
- F. L4
- G. S1



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**11. A 12-year-old-boy is brought to the emergency department because of 2-hour history of shortness of breath. He has a past medical history of asthma. The resident physician conducts auscultation of the thoracic cavity and utilizes a musculoskeletal landmark on the posterior chest wall to listen to lung sounds. Which of the following movements will most likely widen this anatomical landmark?**

- A. Adduction of Scapula
- B. Depression of Scapula
- C. Elevation of scapula
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**12. A 48-year-old man comes to the physician because of a 6-month history of lower back pain. He describes the pain as dull and non-radiating, worsened by prolonged sitting. He does not have any recent trauma, weight loss, numbness, or weakness. His blood pressure is 130/95 mmHg, pulse is 75/min, respirations are 16/min, and temperature is 37°C (98.6°F). An MRI of the lumbar spine is attached. Which of the following embryologic structures most likely gives rise to the structure indicated by the tip of the arrow?**

- A. Neural tube
- B. Notochord
- C. Myotome
- D. Neural crest
- E. Sclerotome
- F. Dermatome



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- A. L2
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**14. A 40-year-old woman is brought to the emergency department by her partner because of 1-hour history of severe neck pain and headache following a fall. Vital signs are within normal limits. Physical examination shows reduced neck mobility. An x-ray of the cervical region is shown; the arrows indicate a fracture. Which of the following arteries is at greatest risk for of in this patient?**

- A. Internal thoracic
- B. Costocervical trunk
- C. Vertebral
- D. Subclavian
- E. Thyrocervical



GAS, Cranio Cervical Junction  
Spinal Cord Injury

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SOM.MKII.BPM1.1.MSK.1.ANA  
T.1043

Describe the different components of a typical vertebra, highlighting the features that are unique to each region.

SOM.MKII.BPM1.1.MSK.1.ANA  
T.1063

Describe the blood supply and venous drainage of the spinal cord and its clinical significance. .

SOM.MKII.BPM1.1.MSK.1.ANAT.1121

Describe and identify (in imaging studies) the anatomical location, mechanism and clinical sequences of a Jefferson fracture, compression fractures of the vertebrae including wedge and Hangman's fracture

**15. A 10-year-old boy is brought to the physician due to difficulty moving his head and neck. His parents report that he has had a short neck and restricted neck movement since birth. His temperature is 36.8°C (98.2°F), pulse is 76/min, respirations are 16/min, blood pressure is 110/72 mm Hg. Physical examination shows a short neck with limited range of motion. There is no scoliosis, but mild asymmetry of the shoulders. Spine X-ray shows fusion of multiple vertebrae. Which of the following is most likely diagnosis?**

- A. Sacralization
- B. Lumbarization
- C. Meningomyelocele
- D. Congenital scoliosis
- E. Klippel-Feil syndrome

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**16. A 50-year-old woman comes to the physician because of a 6-month history of lower back pain. She was involved in a car accident 6 months ago that resulted in her being immobilized for a prolonged period. Physical examination shows significant atrophy of the paraspinal muscles. Which of the following is the most likely functions of the affected structure?**

- A. Extension and flexion of the vertebral column
- B. Flexion and lateral flexion of the vertebral column
- C. Flexion and rotation of the vertebral column
- D. Extension and lateral flexion of the vertebral column
- E. Abduction of the vertebral column and rotation of the vertebral column



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